MA388 Sabermetrics: Lesson 29

Home Run Hitting II

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```r
library(tidyverse)
library(knitr)
library(baseballr)
```

Lesson Objectives

1. Estimate the optimal launch angle for hitting a home run using 2023 Statcast data.
2. Analyze spray angle distributions for home runs by batter handedness.
3. Compute park factors to determine which ballparks favor home run hitting.
4. Fit a random effects model to decompose home run responsibility between pitchers and batters.

```r
# Statcast season window for 2023 (regular + early postseason window as example)
# start_date <- as.Date("2023-04-02")
# end_date   <- as.Date("2023-10-07")
#
# # Break the range into chunks (API is happier with day- or week-sized chunks)
# date_chunks <- seq(start_date, end_date, by = "1 week")
# date_ends   <- pmin(date_chunks + 6, end_date)
#
# # Pull BIP for each chunk
# statcast_2023 <- statcast_search(start_date = date_chunks[1], end_date = date_ends[1])
# for(i in 2:length(date_chunks)){
#   message("Fetching: ", date_chunks[i], " to ", date_chunks[i]+6)
#   Sys.sleep(0.75)  # be polite to the endpoint
#   statcast_2023 <<- bind_rows(statcast_2023,statcast_search(start_date = date_chunks[i], e
# }
#
# statcast_2023 <-
#   map2_dfr(date_chunks, date_ends, ~{
```

```
#   message("Fetching: ", .x, " to ", .y)
#   Sys.sleep(0.75)  # be polite to the endpoint
#   statcast_2023 <<- bind_rows(statcast_2023,statcast_search(start_date = .x, end_date = .y
# })
# # # write_csv(statcast_2023,'statcast_2023.csv')
# #
# bip_2023 <-
#   statcast_2023 |>
#   filter(type=='X') |>
# mutate(hit = events %in% c('single','double','triple','home run'))
#
# write_csv(bip_2023,'statcast_2023_bip.csv')
# #
```

Today, we're going to barely scratch the surface on several aspects of home run hitting (so many more things we could do!). We'll use 2023 Statcast data for everything we do in this lesson.

- What is the optimal launch angle for a home run?
- Do most batters really "pull" the ball when they hit home runs?
- Are some ballparks "hitters' parks?"
- Do some ballparks seem more favorable to either left-handed hitters or right-handed hitters?
- Are home runs a pitcher's failure or a batter's accomplishment?

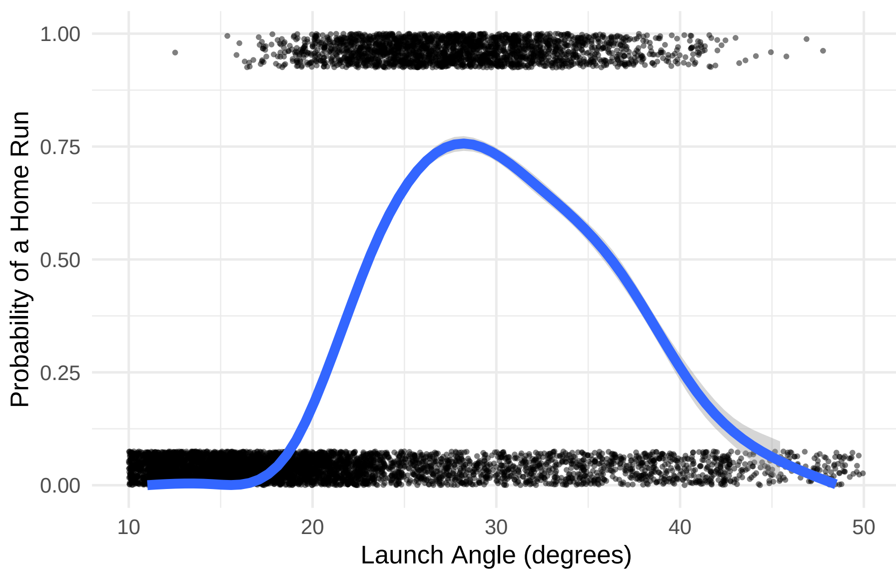
```
library(tidyverse)
library(knitr)

df_23 <- read_csv("statcast_2023_bip.csv") |>
  mutate(HR = events == "home_run")
```

What is the optimal launch angle for a home run?

We know from previous lessons that home runs require both a suitable launch angle and exit velocity. Let's assume you can hit it hard enough, which tends to be 100+ mph for nearly all home runs.

```
df_23 |>
  filter(launch_speed >= 100) |>
  ggplot(aes(x = launch_angle, y = as.numeric(HR))) +
```

```r
  geom_jitter(height = 0.075, width = 2, size = 0.5, alpha = 0.5) +
  geom_smooth(method = "gam", size = 2) +
  scale_y_continuous(
    "Probability of a Home Run",
    limits = c(0, 1)
  ) +
  scale_x_continuous(
    "Launch Angle (degrees)",
    limits = c(10, 50)
  ) +
  theme_minimal()
```



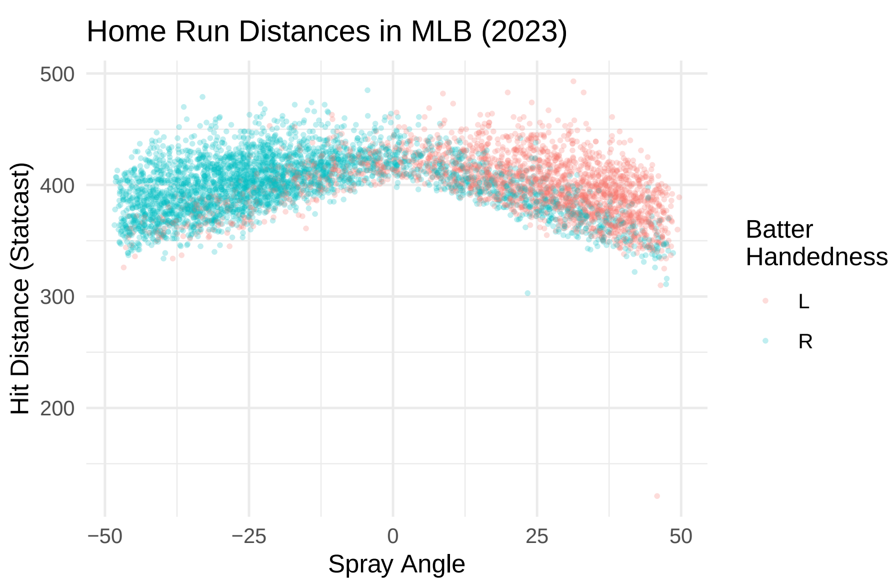
Next Steps:

We've looked at league-wide distributions of launch angles in different years, specifically 2015 and 2017. We didn't see a lot of change across the entire league despite the large increase in home runs, but we might see a different result if we look at a few players whose home run numbers have improved dramatically. Are there individuals whose distribution of launch angles and/or distribution of exit velocities has become significantly more favorable for hitting home runs?

Do most batters really "pull" the ball when they hit home runs?

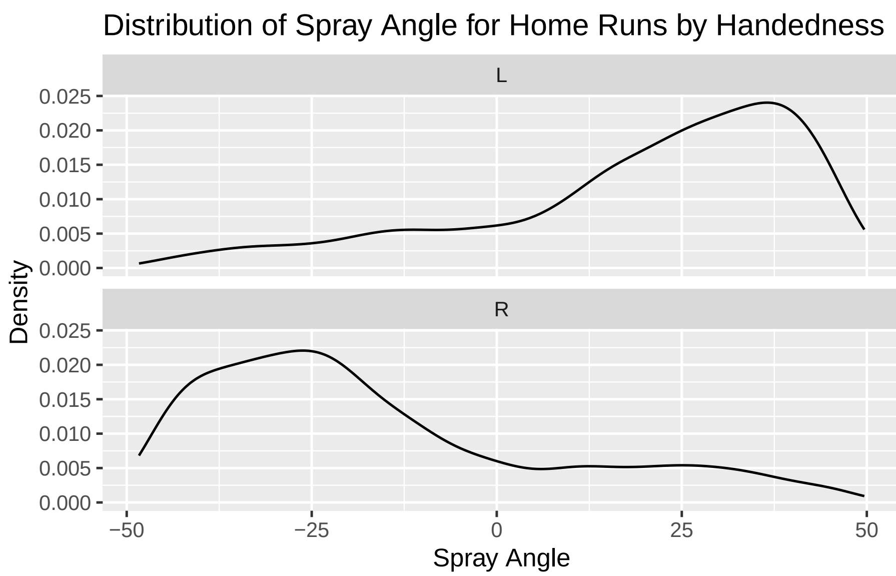
Let's look at spray angle for home runs. We'll need to consider the handedness of the batters as well.

```
df_23 <- df_23 |>
  mutate(
    location_x = 2.5 * (hc_x - 125.42),
    location_y = 2.5 * (198.27 - hc_y),
    spray_angle = atan(location_x / location_y) * 180 / pi
  )
df_23 |>
  filter(events == "home_run") |>
ggplot(aes(spray_angle, hit_distance_sc, color = stand)) +
  geom_point(alpha = 0.25, size = .5) +
  labs(title = "Home Run Distances in MLB (2023)",
       x = "Spray Angle",
       y = "Hit Distance (Statcast)",
       color = "Batter\nHandedness") +
  theme_minimal()
```



Interestingly, there seems to be a relatively sharp cutoff at center field where (1) the dominance of one group of players changes abruptly, and (2) there seem to be relatively few home runs hit by comparison to left and right field.

```r
df_23 |>
  filter(events == "home_run") |>
  ggplot(aes(spray_angle)) +
  geom_density() +
  facet_wrap(vars(stand), ncol = 1) +
  labs(title = "Distribution of Spray Angle for Home Runs by Handedness",
       x = "Spray Angle",
       y = "Density")
```



Next Steps:

It certainly looks like both left-handed and right-handed batters pull the ball for most home runs. Could it be that this is an artifact of differences in ballparks where most home runs for left-handed batters are in ballparks with shorter right-field fences? (…and right-handed batters are succeeding in ballparks with short left-field fences.) Maybe center field is just much deeper and harder to reach.

Do some ballparks seem more favorable to either left-handed hitters or right-handed hitters?

We'll use the "home team" variable in Statcast to help us figure out what percentage of home runs for left-handed and right-handed batters are coming from each ballpark.

```
df_23 |>
  filter(events == "home_run") |>
  group_by(stand, home_team) |>
  summarize(HR = n()) |>
  mutate(percent = HR / sum(HR)) |>
  pivot_wider(names_from = "stand", values_from = c("HR", "percent")) |>
  mutate(percent_diff = percent_L - percent_R) |>
  arrange(-percent_diff) |>
  kable(col.names = c("Ballpark", "HR (L)", "HR (R)", "% of HR (L)",
                      "% of HR (R)", "Diff (%)"),
        digits = 3,
        caption = "MLB ballparks sorted by most favorable to least favorable to left-handed
```

Table 1: MLB ballparks sorted by most favorable to least favorable to left-handed batters (by difference in relative percentages of home runs achieved by left- and right-handed batters).

Ballpark	HR (L)	HR (R)	% of HR (L)	% of HR (R)	Diff (%)
BAL	95	66	0.043	0.021	0.023
BOS	87	85	0.039	0.026	0.013
PIT	71	64	0.032	0.020	0.012
SEA	82	83	0.037	0.026	0.011
CLE	60	57	0.027	0.018	0.009
AZ	78	84	0.035	0.026	0.009
NYM	83	97	0.038	0.030	0.007
ATL	102	129	0.046	0.040	0.006
PHI	94	119	0.043	0.037	0.006
SF	64	78	0.029	0.024	0.005
WSH	75	95	0.034	0.030	0.004
MIN	89	120	0.040	0.037	0.003
MIA	65	87	0.029	0.027	0.002
LAD	95	132	0.043	0.041	0.002
MIL	76	108	0.034	0.034	0.001
STL	73	105	0.033	0.033	0.000
CIN	84	127	0.038	0.040	-0.001
CHC	70	114	0.032	0.036	-0.004
NYY	79	128	0.036	0.040	-0.004
TOR	60	106	0.027	0.033	-0.006
ATH	53	100	0.024	0.031	-0.007
TEX	91	156	0.041	0.049	-0.007
DET	57	109	0.026	0.034	-0.008
COL	65	121	0.029	0.038	-0.008

Ballpark	HR (L)	HR (R)	% of HR (L)	% of HR (R)	Diff (%)
KC	60	114	0.027	0.036	-0.008
LAA	69	131	0.031	0.041	-0.010
SD	54	111	0.025	0.035	-0.010
TB	58	123	0.026	0.038	-0.012
CWS	55	121	0.025	0.038	-0.013
HOU	60	140	0.027	0.044	-0.016

Next Steps:

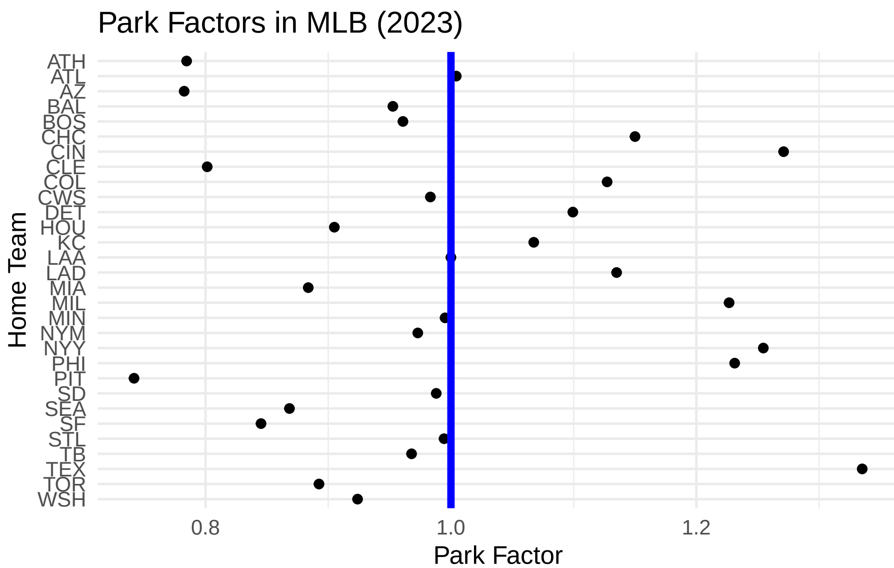
Baltimore certainly seems to stand out as a ballpark that favors left-handed batters. The distances to left and right field are now considerably different after a 2022 renovation extended the left field fence (both in height and depth). (https://www.mlb.com/news/how-2022-camden-yards-changes-will-affect-hitters) We could consider a prior season and see if the left-handed advantage was this strong before the renovation. Next, we'll look at ballparks more generally without any regard for batter handedness. Maybe some ballparks are just more likely to produce home runs due to temperature, altitude, field dimensions, etc.

Are some ballparks "hitters' parks?"

An interesting approach to this is to consider how many home runs are hit by a team and their opponents when the team is at home versus when they are away. Since most baseball teams play each other in roughly the same number of home and away games, this has the effect of seeing how well the same groups of batters do across many different ballparks.

```
df_home <- df_23 |>
  group_by(home_team) |>
  summarize(HR = sum(HR))
df_away <- df_23 |>
  group_by(away_team) |>
  summarize(HR = sum(HR))
park_factor_df <- df_home |>
  inner_join(df_away, join_by(home_team == away_team)) |>
  mutate(park_factor = HR.x / HR.y)
park_factor_df |>
  ggplot(aes(park_factor, home_team)) +
  geom_point() +
  geom_vline(xintercept = 1, color = "blue", size = 1.5) +
  labs(title = "Park Factors in MLB (2023)",
```

```r
       x = "Park Factor",
       y = "Home Team") +
  scale_y_discrete(limits = rev) +
  theme_minimal()
```



Next Steps:

Maybe it's not always about the ballpark dimensions, the weather, etc. Perhaps some teams just have amazing pitchers who don't give up many home runs. Maybe it's the batters that are so good and will hit home runs off any pitcher. Let's now consider the relative effect that individual pitchers and batters have on the likelihood of a home run.

Are home runs a pitcher's failure or a batter's accomplishment?

Here we fit a random effects model with random effects for both the pitcher and batter. The response is whether or not the batter hit a home run. We'll consider the relative size of the standard deviation estimates for pitchers and batters to see if one group or the other seems to be more responsible for home runs.

Our model would look like this:

$$log\left(\frac{\pi_i}{1-\pi_i}\right) = \mu + \beta_j + \gamma_k$$

where π_i is the probability of a home run on pitch i, μ is an overall effect, β_j is the effect due to the jth hitter, and γ_k is the effect due to the kth pitcher. We assume the batter and pitcher effects follow normal distributions with mean 0 and standard deviations σ_b and σ_p, respectively.

```
library(lme4)
fit <- glmer(
  events == "home_run" ~ (1 | pitcher) + (1 | batter),
  data = df_23,
  family = binomial
)
VarCorr(fit)
```

```
 Groups  Name        Std.Dev.
 pitcher (Intercept) 0.19715
 batter  (Intercept) 0.52223
```

We see that the estimate for σ_b is much larger than σ_p, indicating that the variation in home run outcomes is much more attributable to the batters than the pitchers.

Next Steps:

We could make a model that predicts home runs based on pitch location, velocity, movement, etc. Using that model, we could see which pitchers have more or fewer home runs hit against them than expected. We could do the same for batters to see who is making the most out of tough pitches. Think of it like home run expectancy given some pitch variables. Which batters have positive "home runs created" after considering the pitches they saw?